Lab Task: CardioNet – Intelligent System for Heart Disease Prediction

Objective

The objective of this lab is to implement Logistic Regression from scratch and analyze the effects of regularization, overfitting, and underfitting on model performance. Students will gain hands-on experience in building a binary classification model manually, applying L2 regularization, and evaluating bias–variance tradeoff using the Heart Disease dataset.

Dataset

You will be provided with the heart.csv dataset, containing patient attributes such as:

- Age
- Cholesterol level
- Blood pressure
- Maximum heart rate achieved
- Other relevant medical measurements

The dataset also includes a **target variable**:

- 0 → No Heart Disease
- 1 → Heart Disease

Tasks

1. Data Loading and Exploration

- Load the dataset using Pandas.
- Display the dataset's shape, data types, null values, and summary statistics.
- View the first few rows of the dataset to understand its structure.

2. Data Visualization

- Plot the distribution of the target classes.
- Generate a correlation heatmap to visualize relationships between features.
- Plot histograms for each feature to observe data distributions.

3. Data Preprocessing

- Handle missing values using an appropriate imputation strategy.
- Standardize or normalize features if necessary to improve model performance.

4. Model Building

You are NOT allowed to use any machine learning library for model training.

Implement the following manually:

- Sigmoid function
- Binary Cross-Entropy (Log Loss)
- Gradient computation
- Gradient Descent optimization

Train the model using:

- Fixed learning rate
- Fixed number of iterations

4. Model Training Regularization

- Implement **L2 regularization** manually.
- Train the model with different values of the regularization parameter λ (e.g., λ=1 and λ=10).
- Compare model performance with the non-regularized model.
- Discuss how regularization affects:
    - Training accuracy
    - Testing accuracy
    - Model generalization

5. Underfitting Experiment

- Select a subset of features (e.g., age, chol, thalach).
- Train the model using only these features.
- Compare performance with the full-feature model.
- Identify signs of underfitting.

5. Model Evaluation

- Evaluate the models using the following metrics:
    - Accuracy
    - Precision
    - Recall
    - F1-Score
    - ROC-AUC

Submission Requirements

Students must submit:

- Jupyter Notebook (.ipynb)
- Short written analysis (1–2 pages) covering:
    - Overfitting and underfitting
    - Effect of regularization
    - Final conclusions on model selection